Notes

What is a Server?

- If we just look at the word itself, `server` is the one that serves something.
- The system, that is making the request for something, is called the `client`
- `Server` accepts the request from the `client` and sends the response based on the request.

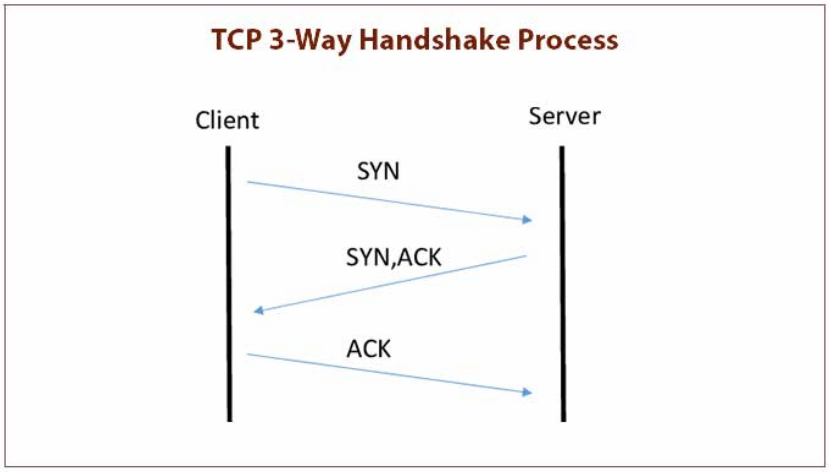
HTTP

- It stands for `Hyper Text Transfer Protocol` .
- Set of Protocol or rules that are required to communicate between client and server.
- **HTTPS:-** It is exactly the same with an added layer of security, it is achieved by SSL(Secure Socket Layer).

3 Way Handshake

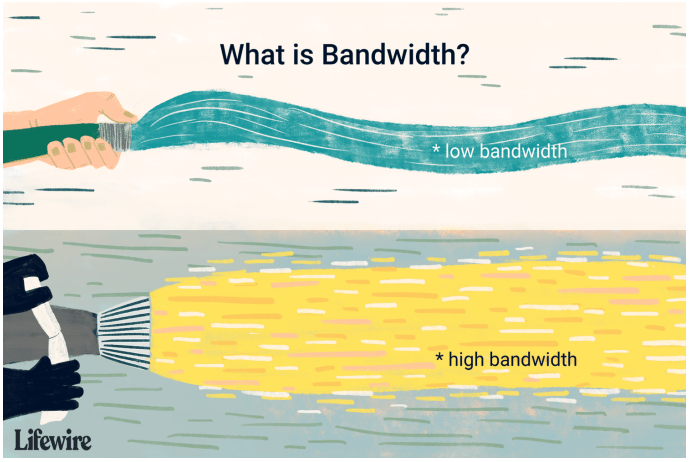
A 3 Way handshake takes place between the client and server, before the actual cycle starts

- First the client expresses the intent to the server.
- Server Acknowledges the client's intent.
- Client will tell what it actually needs.



Bandwidth

- Bandwidth is **the data transfer capacity of a computer network.**



HTTP Verbs/Methods

- **GET:** To read something from the server.
- **PUT⇒Modify:** Will replace the whole thing.
- **PATCH⇒Modify:** Will modify one specific thing in the whole thing.
- **Delete:** To delete something on the server.
- **POST:** To post/add/sent something to the server.
- We can achieve CRUD through this.

Creating First Server

- Create a node project by `npm init -y` .
- Create a file named `index.js` .
- Now for creating the server we can use the inbuilt `http` module of node.

```
const http=require("http")

const server=http.createServer((request,response)=>{
  if(request.url=="/"){
    response.end("Hello")
  } else if(request.url=="/reports"){
    response.end("Here are the reports")
  } else if(request.url=="/data"){
    response.end("Data....")
  }
})

server.listen(4500,()=>{
  console.log("Listening on the port 4500")
})
```

- We have programmed our server to give the response as per the request made.
- Whenever we make any changes in server we have to re run the server.

`.end` VS `.write`

> 💡 **Here we are using `.write`, Now the client will not know that the response has been ended and it will keep on loading the page, that is why we have to use `.end`, so that the client knows that response has been ended.**

Invalid End Point

- What if the user is making a request to an invalid end point, then we have to take care of that as well while programming our server.

```
const http=require("http")

const server=http.createServer((req,res)=>{
  if(req.url=="/"){
    res.end("Hello")
  } else if(req.url=="/reports"){
    res.end("Here are the reports")
  } else if(req.url=="/data"){
    res.end("Data....")
  } else{
    res.end("Invalid End Point")
  }
})

server.listen(4500,()=>{
  console.log("Listening on the port 4500")
})
```

Send Data from a file

- We can also send other things as a response as well.
- Let us try sending data which is inside a file as a response.
- Create a `text.txt` with some dummy data inside it.

```
const http=require("http")
const fs=require("fs")
```

```
const server=http.createServer((req,res)=>{
  if(req.url=="/data"){
    fs.readFile("./text.txt",{encoding:"utf-8"},(err,data)=>{
            if(err){
                res.write("No data\n")
                res.end(err)
            } else {
                res.end(data)
            }
        })
  }
})

server.listen(4500,()=>{
  console.log("Listening on the port 4500")
})
```

> 💡 **The above code will result in getting data as the response of the request is made at `/data` endpoint.**

Headers

- **What are headers?** ⇒ It just gives more information about the request or response.
- We also want to send a header as a response.
- This is just to specify what kind of response we are getting.
- Let's pass a header as well.

```
const http=require("http")
const fs=require("fs")

const server=http.createServer((req,res)=>{
  if(req.url=="/"){
    res.setHeader("Content-type", "text/html") //Header to specify that the reposnse is in HTML form
    res.end("<h1>Hello Guys!!</h1>")
  }
})

server.listen(4500,()=>{
  console.log("Listening on the port 4500")
})
```

Research Work:

💡 **Stream:** https://nodejs.org/dist/latest-v18.x/docs/api/stream.html
Also research about what exactly this is.

HTTP: https://nodejs.org/dist/latest-v18.x/docs/api/stream.html